

Ant colony algorithm for the unsupervised
disambiguation of texts

REFERENCE (Bibliografie)

- [Ant Colony Algorithm for the Unsupervised Word Sense Disambiguation of Texts: Comparison and Evaluation](#) (D. Schwab, J. Goulian, A. Tchechmedjiev, H. Blanchon, in: "Proceedings of COLING 2012", Mumbai, pp. 2389-2404)

- The method refers to knowledge-based unsupervised word sense disambiguation (it is at the border between these two types of WSD)
- The learning process is unsupervised but the algorithm is fed knowledge (it uses extended Lesk, namely knowledge provided by WordNet).
- The local Lesk measure is propagated within an entire text.
- Generally, a global algorithm is a method that allows the propagation of a local measure within an entire text, in order to assign a sense to each word of the text.

- Global disambiguation: analysis is performed at text level. The context window is replaced with the entire text (global algorithm).

Notations:

- w_i = ambiguous word
- m = number of words in the entire text
- $w_{i,j}$ = the most adequate sense of word w_i given the context
- $d(w_{i,j})$ = definition of a sense j of word i

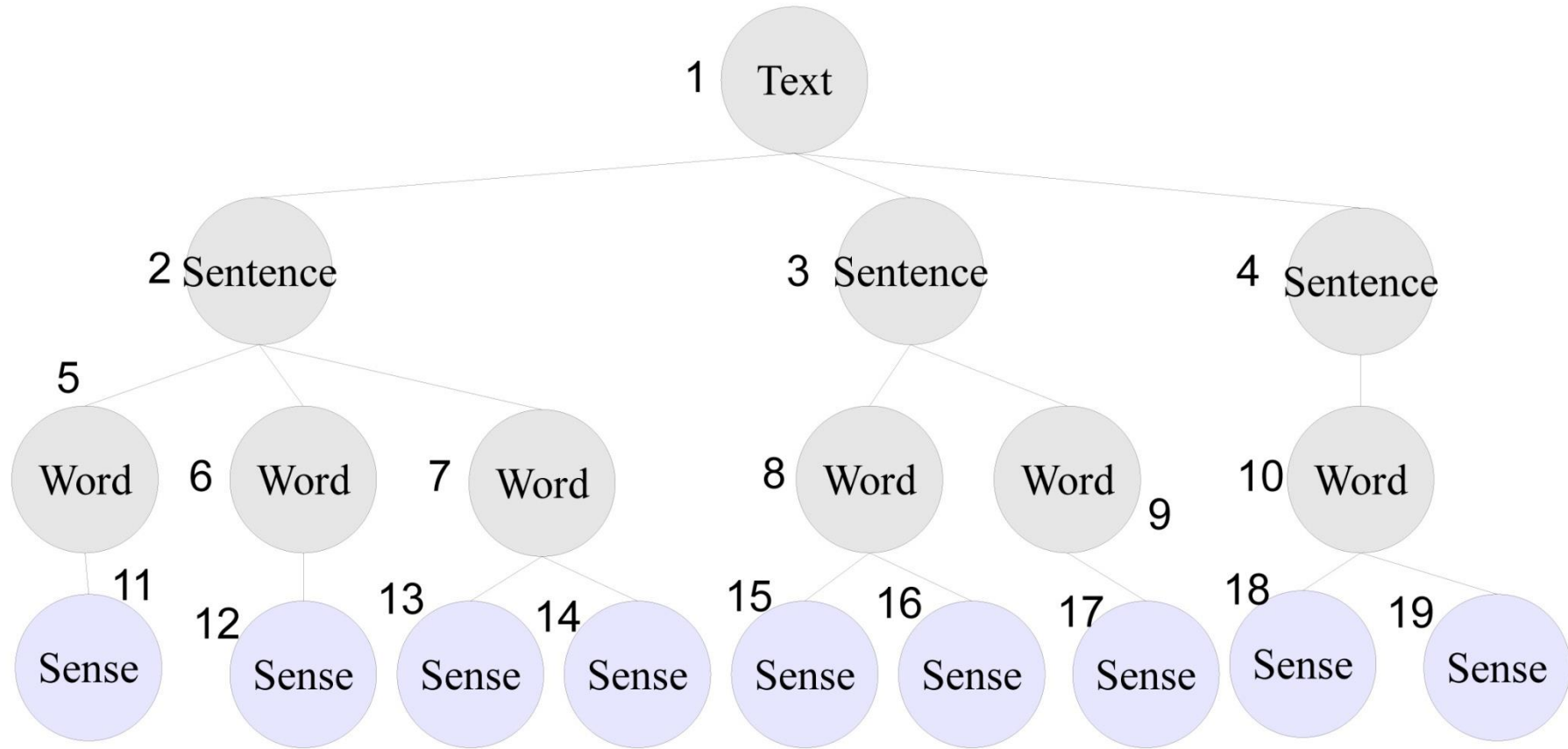
- The search space corresponds to all possible combinations of all senses of all (content) words occurring in the text that is being processed.
- We denote by C a configuration of the problem.
- The representation of a configuration C of the problem consists of a vector of integers such that $j = C[i]$ represents sense j being assigned to word w_i .

Artificial ants

- They were first used for solving “the Traveling Salesman Problem” (Dorigo and Gambardella, 1997)
- The environment (mediul) is usually represented by a graph within which virtual ants explore pheromone trails (dare de feromon - drumuri) created by those ants. (When they move in the graph, the ants leave pheromone trails on the edges they pass through).
- Advantages:
 - A good alternative for solving problems modelled by graphs.
 - Great adaptability to environment changes.

- **Each possible sense of a word is associated to a nest (leaf node).**
- Nests produce ants which move within the graph.
- Ants carry an odour (miros – an array) containing the content words of the definition of the sense corresponding to the ant's mother nest.
- Types of nodes:
 - The mother nest (where the ant is born)
 - An enemy nest (corresponds to another sense of the same word)
 - A potential friend nest (represented by any other nest)
 - A plain node (any node that is not a nest)
- An odour vector (vector de miros) of fixed length, that is initially empty, is associated to each plain node.

- **Example of a graph representing the environment:**



The root of the tree corresponding to the graph is the entire text.

For instance, the word corresponding to node 7 is polysemous, thus ambiguous.

- **Ant movement** depends on:
 - the scores computed by the local algorithm;
 - the presence of energy;
 - the movement of other ants (when passing by, ants leave a trail of pheromone which, in time, fades away);
 - the odour vectors of the nodes (ants deposit a part of their odour on the nodes they go through).
- When an ant arrives on a nest of another term (that corresponds to a sense), it can either continue its exploration or, depending on the score between this nest and its mother nest, it can decide to build a bridge between them and to follow it home.

- Bridges behave like normal edges except that, if at any given time, the concentration of pheromone reaches 0, the bridge collapses. Ants will favor following bridges between more closely related senses.
- The more closely related the senses of the nests are, the more a bridge between them will contribute to their mutual reinforcement and to the sharing of resources between them (thus forming *meta-nests*).
- The bridges between more distant senses will tend to fade away.
- We are thus able to build “interpretative paths”, through emergent behavior (comportament emergent), and to suppress the need to use a complete graph that includes all the links between the senses from the start. – constructie de “drumuri interpretative”

- The execution of the algorithm is a potentially infinite succession of **cycles**. After each cycle, the state of the environment can be observed and used to generate a solution.
- **A cycle is composed of the following actions:**
 - eliminate dead ants and bridges with no pheromone;
 - for each nest, potentially produce an ant;
 - for each ant: determine its mode (energy search or return); make it move; potentially create an interpretative bridge;
 - update the environment (energy levels of nodes, pheromone and odour vectors).

Main notations for the algorithm:

F_A – nest that corresponds to sense A

f_A – ant born in nest F_A

$V(X)$ – odour vector associated with X (ant or node)

$E(X)$ – energy on/carried by X (ant or node)

$\text{Eval}_f(N)$ – evaluation of a node N by an ant f

$\text{Eval}_f(A)$ – evaluation of an edge A (quantity of pheromone) by an ant f

$\varphi_{(t/c)}(A)$ – quantity of pheromone on edge A at given moment t or cycle c

Parameters for the algorithm:

Notation	Description
E_a	Energy taken by an ant when it arrives on a node
$E_{\max}$	Maximum quantity of energy an ant can carry
δ	Evaporation rate of the pheromone between two cycles
E_0	Initial quantity of energy on each node
ω	Ant life-span
L_v	Odour vector length
δ_v	Percentage of the odour vector components (words) deposited by an ant when it arrives on a node
C_{ac}	Number of cycles of the simulation

1. Ant production, death and energy model

Initially, we assign a fixed quantity of energy E_0 to each node of the environment. At the beginning of each cycle, each nest node N has an opportunity to produce an ant A using 1 unit of energy, with a probability $P(N_A)$. The literature defines this probability as the following function

$$P(N_A) = \frac{\arctan(E(N))}{\pi} + 0.5$$

where $E(N)$ is the energy of node N .

When created, an ant has a lifespan of ω cycles. When the life of an ant reaches zero, the ant is deleted at the beginning of the next cycle and the energy it carried is deposited on the node where it died. By thus doing, we ensure that **the global energy equilibrium of the system** is preserved, which plays a fundamental role in the convergence (monopolization of resources by certain nests) to a solution.

2. Ant movements

- The ants' movements are random, but influenced by the environment.
- When an ant is on a node, it assigns a transition probability to the edges leading to all neighbouring nodes. The probability to cross through an edge A_j in order to reach a node N_i is:

$$P(N_i, A_j) = \frac{Eval_f(N_i, A_j)}{\sum_{k=1, l=1}^{k=n, l=m} Eval_f(N_k, A_l)},$$

where $Eval_f(N, A) = Eval_f(N) + Eval_f(A)$ is the evaluation function of a node N when coming from an edge A .

A newborn ant **seeks food** in the following way:

- a) It is attracted by the nodes which carry the most energy:

$$\text{Eval}_f(N) = \frac{E(N)}{\sum_{i=0}^m E(N_i)}$$

- b) It avoids to go through edges with a lot of pheromone:

$$\text{Eval}_f(A) = 1 - \varphi_t(A),$$

where $\varphi_t(A)$ is the quantity of pheromone of edge A at moment t, in order to favor a greater exploration of the search space.

- The ant collects as much energy (food) as possible, until it decides to bring it back home (**return mode**) with the probability:

$$P(\text{return}) = \frac{E(f)}{E_{\max}},$$

where $E(f)$ is the energy that ant f has at the current moment and $E_{\max}$ is the maximum quantity of energy that an ant can carry.

Remark: When an ant f reaches its maximum transportation capacity, the probability of it returning home is 1.

- When an ant decides to enter the **return mode**, it moves following (statistically) the edges that contain the most pheromone,

$Eval_f(A) = \varphi_t(A)$, where:

- $Eval_f(A)$ represents the evaluation of edge A by ant f (quantity of pheromone)
- $\varphi_t(A)$ represents the quantity of pheromone of edge A at moment t.
- Also, the edges must lead to nodes with an odor close to their own (that of the ant) ,

$$Eval_f(N) = \frac{ExtLesk(V(N), V(f_A))}{\sum_{i=1}^k ExtLesk(V(N_i), V(f_A))}, \text{ where:}$$

- $V(N)$ – odor vector associated to node N
- $V(f_A)$ – odor vector associated to the ant born in node F_A
- $ExtLesk()$ – the Extended Lesk algorithm

Remark: The Extended Lesk algorithm is applied here in connection with two odor vectors.

3. Creation, deletion and types of bridges

When an ant arrives on a node adjacent to a potential friend nest (namely one that corresponds to a sense of a word) it has to decide between taking any of the possible paths or to go on that nest node. As such, we are dealing with a particular case of the ant path selection algorithm, namely the case with:

$\text{Eval}_f(A)=0$ (the pheromone on the edge is ignored)

The only difference is that if the ant chooses to go on the potential friend nest, a bridge between that nest and the ant's home nest is built and the ant follows it to go home.

Bridges behave like regular edges, except that if the concentration of pheromone on them reaches 0, they collapse and are removed.

4. The pheromone model

- ❑ When they move in the graph, ants leave pheromone trails on the edges they pass through. Ants have two types of behaviors:
 - they are looking to gather energy
 - or
 - they are looking to return to their mother nest
- ❑ The pheromone density on an edge influences the movements of the ants : they prefer to avoid edges with a lot of pheromone when they are seeking energy and to follow them when they want to bring the energy back to their mother nest.

- When passing on an edge A , ants leave a trail by depositing a quantity of pheromone $\theta \in \mathbb{R}^+$ such that

$$\varphi_{t+1}(A) = \varphi_t(A) + \theta$$

- Furthermore, at each cycle, there is a slight (linear) evaporation of pheromones (penalizing little frequented paths):

$$\varphi_{t+1}(A) = \varphi_t(A) * (1 - \delta)$$

where δ is the pheromone evaporation rate.

5. Odour

Definition: *The odor of a nest is a numerical sense vector corresponding to the definition of the sense associated to that nest.* This numerical sense vector is determined as follows:

- We apply Extended Lesk, slightly modified, such that each word occurring in any of the glosses is indexed by a unique integer number. The modification resides in the fact that multiple words overlapping do not get a higher score. Individual words are treated individually (overlapping of type “bag of words”). This is done in order to decrease complexity from $O(m*n)$ to $O(m)$, $m>n$, where m and n are the lengths of the two definitions that are being compared. By means of this indexing, the **entire definition** receives a score represented by a vector.

Example:

The definition: “Some kind of evergreen tree”

It was shown that:

“same” is indexed by 123

“kind” is indexed by 14

“evergreen” is indexed by 34

“tree” is indexed by 90

Then the indexed representation of the entire definition is:

{14,34,90,123} (Remark: the vector is sorted)

- All ants **born in the same nest** have the same odor vector. When an ant arrives on a common node N , it deposits some of the components of its odor vector (following a uniform distribution), which will be added to or will replace the existing component of the node's vector $V(N)$.
- The odor of nest nodes, on the other hand, is never modified.
- This mechanism allows ants to find their way back to their mother nest. Indeed, the closer a node is to a given nest, the more ants from that nest will have passed through and deposited odor components.

- Thus, the odor of that node will reflect its nest neighborhood and allow ants to find their way by computing the score between their odor (that of their mother nest) and the surrounding nodes and by choosing to go on the node yielding the highest score.
- This process leaves some room for error (such as an ant arriving on a nest other than its own), which is beneficial as it leads ants to build more bridges.

Global evaluation

- At the end of each cycle, we build the current problem configuration from the graph: for each word, we take the sense corresponding to the nest with the highest quantity of energy.
- Subsequently, we compute the global score of the configuration:
 - The score of a selected sense of a word can be expressed as the sum of the local scores between that sense and all the other senses selected for all the other words of a context (sentence).
 - For the global score of the configuration we add the scores for all selected senses corresponding to all words of the text:

$$Score(C) = \sum_{i=1}^m \sum_{j=1}^m ExtLesk(w_{i,C[i]}, w_{j,C[j]})$$

$j=C[i]$ is the selected sense j for word w_i

Remark: Here we see that we are not working with a context window, but with the entire text.

- Complexity:
 $O(m^2)$, where m = the number of words in the entire text.
- Over the execution of the algorithm we keep the configuration with the highest absolute score, which will be used at the end to generate the solution.

Main parameter values:

Notation	Description	Value
E_a	Energy taken by an ant when it arrives on a node	1 – 30
$E_{\max}$	Maximum quantity of energy an ant can carry	1 – 60
δ	Evaporation rate of the pheromone between two cycles	0.0 – 1.0
E_0	Initial quantity of energy on each node	5 – 60
ω	Ant life-span	1 – 30 (cycles)
L_v	Odor vector length	20 – 200
δ_v	Percentage of the odor vector components (words) deposited by an ant when it arrives on a node	0 – 100%
c_{ac}	Number of cycles of the simulation	1 – 500

Empirical evaluation

(Evaluare empirica)

- ❑ **The optimal parameter values can not be determined in an analytical way and have been estimated experimentally.**
- ❑ **The performed experiment** consisted of the annotation of 2269 words with one of their possible senses given by WordNet. The average degree of polysemy was 6.19. The corpus of the competition SemEval 2007 was used.
- ✓ As a result of these tests, the best parameter values that were found are:

$$\omega = 25, E_a=16, E_{\max}=56, E_0=30, \delta_v=0.9, \delta=0.9, L_v=100$$

□ Tests performed in 2013 have led to the following parameter values:

• For English

$$\omega = 26, E_a=14, E_{\max}=3, E_0=34, \delta_v=0.9775, \delta=0.3577, L_v=25$$

• For French

$$\omega = 19, E_a=9, E_{\max}=3, E_0=32, \delta_v=0.9775, \delta=0.3577, L_v=25$$

Test results

- Obtained accuracy (F1) on SemEval-2007 data: **76.41%**
- The algorithm was superior to the best “state of the art” unsupervised system and to the weakest supervised system.

The type of algorithm:

- It performs global disambiguation (the entire text represents the disambiguation context).
- It is of type knowledge based unsupervised (the learning process is unsupervised; knowledge coming from WordNet is provided to the algorithm).